DSC-209R Project-1

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This visualization examines the relationship between affordability and FPro scores between baking products sold at Walmart, Target, and Whole Foods. The analysis focuses on determining which retailer offers the greatest share of products that are less processed and budget friendly. The intended audience is budget-conscious families seeking ways to reduce their consumption of ultra-processed foods without increasing grocery costs. The primary finding is that Whole Foods, despite its reputation as a more expensive option, contains the highest share of baking products that also meet the criteria of having an FPro score below 0.5 and a price per ounce at or below the median price of all baking items in the dataset ($0.25/oz). Further insights can be seen in Walmart providing the most budget friendly options overall, while Target contains the highest proportion of ultra-processed baking products. The baking category was chosen because it contains a particularly high share of ultra-processed items overall, making it difficult for consumers to find affordable, less-processed items without data driven insights.

Several visual encoding techniques were implemented to communicate the primary takeaway. The percentage based stacked horizontal bar chart was selected because it easily displays expressiveness of the data. The position and length allow one to accurately compare the proportion of baking products in each store. The y-axis identifies the retailers, while the x-axis represents the percentage of baking products in each classification. Color is used to show the product classification, which is a combination of processing level and affordability. Green represents budget friendly and less processed products, purple represents budget friendly and ultra processed products, yellow represents more expensive and less processed products, and red represents more expensive and ultra-processed products. Color encoding green and red for the polar opposites was chosen because it leverages common pre-attentive associations; viewers are likely to interpret green as positive and red as negative. Yellow and purple were selected as intermediate categories that represent products meeting only one of the two desired criteria. Yellow indicates products that are less processed but more expensive, while purple indicates products that are budget friendly but ultra processed. The legend was drawn as a matrix to easily communicate the color encoding without overly relying on text usage. The legend was designed with inspiration from Nigel Holmes', who argued that visual aides can present long-term lasting memories of a data visualization. I made the color encoding in the legend appear as rolling pins, keeping with the baking theme while also tying back to the underlying data. Other Holmes' inspired elements in the visualization are the inclusion of a loaf of bread as a transparency in the lower left corner, and a muffin transparency behind the key findings takeaway summary. The chart was created directly in python using Bokeh plots. As part of that the sub-title I created describes the primary filters that were applied to the visualization.

The title was chosen to use word play and draw a contrast between our product classifications. The constant use of the letter 'B' creates an alliteration that is intended to be memorable and rhythmic. The subtitle is the conclusion that we reached in our analysis. At the top of the chart I print the basic filters that were applied to the data. It specifies that we are only focusing on the baking category, splitting price by the median value of all items (per ounce), and splitting FPro on 0.5. As not all viewers will understand what FPro refers to, I included a short explanation beneath the chart. In chart annotation boxes were added to highlight key insights that viewers might not immediately calculate themselves, such as Walmart's large share of budget friendly baking products, Whole Foods advantage in budget friendly and less processed products, and targets overall ultra processed availability. Additionally, I used further pre-attentive attributes to highlight important findings in the lower right takeaway section. Font size, bold, and italics are used to make specific words pop and draw the viewers attention towards the highlights of the analysis. Source is also included as a citation mechanism for the visualization.

Several data transformations were made to display the relationship between affordability and processing level. The simplest filter that was applied narrowed the data set to only focus on the baking category as pre-analysis showed it to display a high level of ultra processed items. The second transformation was to look at the quantity of products as a Quantitative ratio (percentage) as opposed to raw counts. Raw counts could potentially bias the data and falsely inflate larger stores capable of having more shelf space. Viewing percentages provides a balanced assessment per retailer. Next, to determine a budget friendly option verses a pricier option I used the median value per-ounce of all items. Raw counts could inflate the measurements and not provide the viewer with an accurate understanding of how much product they received for the price. To represent the per-ounce price, I divided baking_price by package_weight and converted from grams to ounces (as you typically see on packaging labels in store). Processing level is calculated by looking at the min and max FPro scores for each classifications. 0.5 is the separable threshold between class 0 (the lowest classification) and class 3 (the highest classification). No class 0 score is higher then 0.499843 and no class 3 score is lower than 0.500005, making 0.5 a fair threshold to use for separability of high and low processing levels.